CIVE 202 – Civil Engineering Analysis II

Project #5 Report

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Introduction/Backgroud Info

The Nebraska Department of Transportation (NDOT) is currently transitioning its asphalt mixture approval process to a Balanced Mix Design (BMD) framework. Unlike traditional asphalt mix evaluation methods that rely primarily on volumetric properties, the BMD approach incorporates performance testing methods that evaluate cracking resistance and rutting resistance. Currently, these calculations are run manually through complex excel sheets. The process is unnecessary, lengthy, and difficult. To improve calculation time and accuracy, NDOT has tasked us with the following. The purpose of this project was to design and develop a modular Python analysis suite that automates four primary pavement mixture evaluation tasks:

- Aggregate Gradation Analysis
- Volumetric Properties Calculations
- IDEAL-CT Cracking Analysis
- Hamburg Wheel Tracking Test (HWTT) Rutting Analysis

Objectives

- Develop Python-based program capable of running NDOT calculations.
- Process raw data from the excel files
- Calculate aggregate gradation and compare to NDOT standards
- Computing volumetric properties
- Automate IDEAL-CT cracking calculations including CTIndex analysis.
- Analyze Hamburg Wheel Tracking Test data to identify rutting behavior and stripping
- Generating readable engineering plots
- Provide more accurate answers with less error.

Scope of Work

Team members: Nicholas Bennet, Eli McNeely, Gavin Rocha

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Client: NDOT

Submitted to: Civil and Environmental Engineering Department Facility

Project Goals

The goal of this project is to automate mixed pavement design calculations using Python Programming. This is to replace the manual process of using the excel spreadsheet to make the calculations yourself. The work is split into four different modules, each being required to,

- Read raw data from excel
- Perform correct calculations
- Generate graphs and plots
- Produce summary outputs
- Reduce manual requirements for calculations.

Project Tasks

Task 1 – Aggregate Gradation Automator

The first task involved developing a module capable of reading aggregate sieve analysis data and blending percentages from Excel spreadsheets. The module computes the combined aggregate gradation curve and compares the resulting mixture against NDOT specification limits. The module also generates a plot of sieve size versus percent passing, including:

- Individual aggregate curves
- Combined gradation curve
- NDOT minimum specification limits
- NDOT maximum specification limits
- Shaded specification control band

Task 2 – Volumetric Properties Calculator

The second task involved creating a volumetric analysis module capable of computing asphalt mixture volumetric properties using laboratory measurements. The module reads the required measurements from Excel spreadsheets and automatically computes:

- Maximum specific gravity (Gmm)

- Bulk specific gravity (Gmb)
- Air void content (Va)

Task 3 – IDEAL-CT Cracking Analyzer

The third task focused on automating the IDEAL-CT cracking analysis procedure. The module reads load versus displacement laboratory data and computes the CTIndex for each asphalt mixture sample. The analysis required the software to:

- Calculate the area under the load-displacement curve
- Determine peak load values
- Locate the 75% post-peak load point
- Calculate the post-peak slope
- Compute fracture energy and CTIndex values

Task 4 – Rutting Analyzer (HWTT)

The final task involved developing a Hamburg Wheel Tracking Test (HWTT) rutting analysis module. The module reads wheel cycle versus rut depth data and evaluates rutting performance for asphalt mixtures. The software automatically:

- Detects stripping inflection points (SIP)
- Identifies rutting failure cycles
- Evaluates pass/fail criteria
- Generates rut depth versus wheel cycle plots

Data Source

The datasets used for this project were provided through the AutoBMD.xlsx workbook supplied as part of the pavement mixture design project. The workbook contained the raw laboratory datasets required to complete all four pavement performance analysis modules. These datasets included aggregate gradation data, volumetric measurements, IDEAL-CT cracking data, and Hamburg Wheel Tracking Test (HWTT) rutting data. The spreadsheet served as the primary input source for the automated Python software.

Data Extraction and Organization

The data was extracted from the xlsx file provided to us and was analyzed through the use of Jupyter notebook. The data was then processed into Data frames that could be processed through Jupyter. After the data was imported, each sheet was organized according to the module it supported. Each value was assigned based on its use.

Data Visualization

Data visualization was used throughout the project to display engineering results in a clear and organized format. The matplotlib and seaborn libraries were used to generate plots for each analysis module. The python code created gradation curves, volumetric summary plots, load versus displacement graphs, and rut depth versus wheel cycle plots. These visualizations helped compare results, identify trends, and evaluate asphalt mixture performance more efficiently.

Results and Discussion

The following conclusion was reached after creation and performance of our assigned python code.

Final Results

This paper shows the results of a Python programming demonstration to help solve pavement mixture design problems and make engineering workflows more productive. The software created could read, analyze, calculate, create visuals, and summarize results from different asphalt performance tests laboratories.

The program was designed in such a way that each part of the analysis could run separately and also follow the main software workflow consistently. By automating calculation tasks directly from the input data, the need for manual spreadsheet work was eliminated, the efficiency was increased, and the possibility of human mistake during the data handling was reduced.

In general, this project has established that scientific computing tools like pandas NumPy matplotlib, and SciPy are not only good for transportation engineering but also can greatly enhance the speed and accuracy of pavement performance analyses.